Fuel Control Unit Push Rod End – Inspection and Lubrication

1. General

The Fuel Control Unit (FCU) push rod assembly connected to the FCU power lever contains a rod end bearing at each end; this allows for the free motion of the linkage and resolves any misalignment between the ends.

The Pre Mod 6/2347 FCU push rod assemblies use rod end bearings (ref P/N DSC4–3A) that have an inner race with a fabric-PTFE liner. The Post Mod 6/2347 FCU push rod assemblies use rod end bearings (ref P/N VSC30–3A) that are fabricated with a metal inner race and a dry film lubricant.

Note Mod 6/2347 FCU push rod assembly bearings are required to be lubricated in all environments.

The Post Mod 6/2484 FCU push rod assemblies use rod end bearings (ref P/N VSC33–3A) that have a brass liner.

2. Inspection of Fuel Control Unit Rod End Bearings

WARNING THE PROCEDURE USES MATERIAL THAT MAY BE DANGEROUS TO THE PERSONNEL'S HEALTH AND/OR THE ENVIRONMENT. SUFFICIENT PRECAUTIONS MUST BE TAKEN AT ANY STEP OF USE. REFER TO THE MATERIAL SAFETY DATA SHEET AND ANY LOCAL REGULATIONS REGARDING EXPOSURE, USE, AND DISPOSAL PRIOR TO COMMENCING WORK.

A. Inspection of Fuel Control Unit Rod End Bearings

1) Switch OFF aircraft electrical power (refer to 24-00-00, Electrical Power – General Data).

2) Lower the engine cowlings.

3) Remove panel to gain access to the Fuel Control Unit.

4) Use a fine tip permanent marker to draw a line across the edge of the serrated washer and the applicable mating parts to ensure that the washer to lever matching remains undisturbed.

Note Slotted adjustment points utilize serrated washers. Ensure serrated washers, where applicable, are properly marked for alignment with the mating parts before disconnecting.

5) Disconnect the FCU control rod from the FCU arm; remove the cotter pins, nuts, bolts and washers.

6) Clean each FCU rod end bearing in accordance with Para 2.B.

7) Identify the rod end bearing type as per Table 1 — Lubrication of Engine Control Rod Ends.

8) Inspect each rod end bearing ball:

a) Manually turn the ball and feel for signs of roughness or excessive drag in rotation. The bearing ball must be easily movable with the fingers. If the rod end bearing ball does not move easily and smoothly, replace.

b) Check the rod end bearing ball for excessive side and radial play. Replace rod ends with excessive side or radial play.

c) Examine the rod end housing and bearing ball for corrosion. Replace rod ends with corrosion on the bearing ball. Remove light corrosion on the housing only with a coarse weave cloth or fibre brush.

Note If light corrosion was removed from the rod end housing, clean rod end bearing in accordance with Para 2.B.

d) Examine the rod end bearing for damage, including:

- excessive scratching
- scoring
- cracks
- flaked plating
- fretting against the edges of the race

e) Examine the rod end race for damage, including evidence of:

- excessive wear as indicated by roughness, or excessive axial or radial play
- liner cracking (fabric-PTFE lined rod ends only)
- detached liner as indicated by the liner being loose or shifted to one side (fabric-PTFE lined rod ends only)
- evidence of corrosion between the housing and the liner or dry-film lubricating coating
- sharp protrusions and/or chips

9) Replace rod ends if damage is observed in Para 2.A.8).

10) Lubricate rod end bearings using the lubricant specified in Table 1 — Lubrication of Engine Control Rod Ends and in accordance with Para 3.

11) Re-install the FCU control rod; install the cotter pins, nuts, bolts and washers. Replace damaged or corroded hardware.

Note Ensure that the marked position of each serrated washer and mating part has remained in alignment. If alignment is not maintained or cannot be confirmed, check rigging as per 76-10-00, Power Control – Maintenance Practices.

12) Check free movement of the linkage.

13) With power levers at IDLE position.

a) Confirm that the FCU power lever is in the idle position (against idle stop).

b) Push FCU push rod forward and release.

c) FCU power lever should immediately return to idle position.

14) Reinstall panel.

15) Close engine cowls.

16) Do an engine run in accordance with 71-00-00, Power Plant (Mod 6/2012) – General Data.

B. Cleaning

1) Materials and Equipment

a) Materials

- Cleaning solvent DS108, Diestone DLS, or Diestone HFP

b) Equipment

- Cloth, coarse weave
- Cloth, lint-free

2) Safety Precautions

a) Sufficient ventilation must be provided when using solvent in confined areas. Wear protective respiratory equipment when applying solvent.

b) Wear protective gloves to avoid skin contact with solvents. If contact occurs, wash the area thoroughly with soap and water. Wash or discard contaminated clothing as recommended by the Material Safety Data Sheet (MSDS).

c) Refer to the applicable manufacturer's MSDS for specific safety data on any of the materials specified herein.

3) Cleaning Procedure

CAUTION **EXCESSIVE SOLVENT MAY DAMAGE OR LOOSEN FABRIC-PTFE LINER OR DRY FILM SELF-LUBRICATING COATING. DO NOT USE EXCESSIVE SOLVENT TO CLEAN ROD END BALL. DO NOT SOAK ROD END.**

CAUTION **WATER MAY DAMAGE, SWELL OR LOOSEN FABRIC-PTFE LINER. DO NOT USE WATER TO CLEAN ROD END BALL. DO NOT SOAK ROD END.**

a) Clean the rod end housing and/or ball using a clean, dry cloth. Remove all residual lubricant, dust and dirt.

b) When necessary, clean with a lint-free cloth lightly dampened with cleaning solvent in accordance with Para 2.B.1).a).

3. Lubrication

Lubrication of all rod ends is required after inspection and, on condition, after wash-out of the engine externals in the area of the power control linkage.

WARNING **ON DRY FILM COATED ROD ENDS, USE ONLY LUBRICANTS SPECIFIED IN Table 1 — Lubrication of Engine Control Rod Ends. DAMAGE TO THE COATING AND/OR ROD END BINDING MAY RESULT.**

CAUTION **DO NOT USE LUBRICANTS ON FABRIC-PTFE LINED ROD ENDS OTHER THAN THOSE SPECIFIED IN Table 1 — Lubrication of Engine Control Rod Ends. DAMAGE TO THE LINER AND/OR EXCESSIVE BREAK-OUT TORQUE MAY RESULT.**

Table 1 Lubrication of Engine Control Rod Ends

Rod End Type	Liner or Coating Type	Liner or Coating Appearance	Lubrication Requirement	
Pre Mod 6/2347	Fabric-PTFE Liner	Amber to dark orange bearing race liner. Edge visible around the ball.	LUBRICATE	MIL-PRF-7870 (Royco 363) OR MIL-PRF-81322 (AeroShell Grease 22)
Post Mod 6/2347	Dry Film Coating	Dark grey to black coating, applied to the inside of the bearing race. Note The dry film lubricant layer is thin, with no edge visible. Rotate the ball and use a bright light to view the inside of the race.	LUBRICATE	
Post Mod 6/2484	Brass Liner	Metallic brass colour.	LUBRICATE	MIL-PRF-23827 TYPE II (AeroShell Grease 7)

Note If there is difficulty identifying the rod end type contact Viking Technical Support.